

pH and the pH Scale

Answer key with worked solutions

A. Understanding

1. Classify.

2.5 acidic; 7.0 neutral; 9.8 basic; 6.9 acidic (just below 7).

2. pH 2 vs pH 5.

A difference of 3 pH units = $10^3 = 1000$ times greater.

B. Strong acids

3. 0.0010 M HCl.

$[\text{H}_3\text{O}^+] = 0.0010 \text{ M}$; $\text{pH} = -\log(0.0010) = 3.00$

4. 0.25 M HNO_3 .

$[\text{H}_3\text{O}^+] = 0.25 \text{ M}$; $\text{pH} = -\log(0.25) = 0.60$

C. Strong bases

5. 0.010 M NaOH.

$[\text{OH}^-] = 0.010 \text{ M}$; $\text{pOH} = 2.00$; $\text{pH} = 14.00 - 2.00 = 12.00$

6. 0.0050 M $\text{Ba}(\text{OH})_2$.

Two OH^- per formula unit: $[\text{OH}^-] = 2 \times 0.0050 \text{ M} = 0.010 \text{ M}$; $\text{pOH} = 2.00$; $\text{pH} = 12.00$

D. Concentration from pH

7. $[\text{H}_3\text{O}^+]$ from pH.

(a) $10^{-3.20} = 6.3 \times 10^{-4} \text{ M}$ (b) $10^{-11.50} = 3.2 \times 10^{-12} \text{ M}$

8. $[\text{OH}^-]$ at pH 5.00.

$$\text{pOH} = 14.00 - 5.00 = 9.00, \text{ so } [\text{OH}^-] = 10^{-9.00} = \mathbf{1.0 \times 10^{-9} \text{ M}}$$

E. Challenge

9. 100-fold dilution of 0.10 M HCl.

New concentration = $0.10 \text{ M} \div 100 = 0.0010 \text{ M}$, so $\text{pH} = -\log(0.0010) = \mathbf{3.00}$. The pH rises by **2 units** (from 1.00), one unit for each factor of 10.